

BPHO ROUND 0 · MARKSCHEME

Drill bank section 3

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.*

Quick key

Q1 C	Q2 D	Q3 B	Q4 A	Q5 E
Q6 C	Q7 A	Q8 B	Q9 E	Q10 D
Q11 C	Q12 A	Q13 E	Q14 B	Q15 D
Q16 A	Q17 C	Q18 E	Q19 B	Q20 D
Q21 A	Q22 C	Q23 B	Q24 E	Q25 D

Worked solutions

Q1 · A loaded beam on two supports: the reaction at one end — **Answer C**

What is being tested. Whether you can choose the pivot that makes the algebra shortest. Nothing here needs a calculator: every number divides cleanly once the pivot is right.

Step 1 — where does the beam's weight act? The beam is uniform, so its weight behaves as though it were all concentrated at the midpoint. The midpoint is 3.0 m from A and 3.0 m from B. This is the fact the whole question rests on; if the beam were not uniform, nothing could be done without more information.

Step 2 — choose the pivot. Two forces are unknown, the reactions at A and at B. Taking moments about B gives the reaction at B a moment arm of zero, so it disappears. One equation, one unknown:

$$\text{moment about B: } F_A \times 6.0 = W_{\text{beam}} \times 3.0 + W_{\text{load}} \times (\text{distance of the load from B})$$

Step 3 — the load's moment arm. The load is 2.0 m from A, and the beam is 6.0 m long, so its distance from B is $6.0 - 2.0 = 4.0$ m. Getting this arm right is the whole of the question's difficulty: the load is near A, and a support takes more of a load that is near it, which is the physical check on the answer.

Step 4 — put the numbers in.

$$\begin{aligned}
 F_A \times 6.0 &= 240 \times 3.0 + 360 \times 4.0 \\
 &= 720 + 1440 \\
 &= 2160 \\
 F_A &= 2160/6.0 = 360 \text{ N}
 \end{aligned}$$

So **Answer: C.**

Step 5 — check it two ways. First, the forces must add up: the total load is $240 + 360 = 600 \text{ N}$, so the other support carries $600 - 360 = 240 \text{ N}$. Second, take moments about A with that value: $240 \times 3.0 + 360 \times 2.0 = 720 + 720 = 1440$, and the reaction at B gives $240 \times 6.0 = 1440$. The two agree, so 360 N is right.

Step 6 — does the size make sense? The load of 360 N sits 2.0 m from A, which is well inside the half of the beam nearest A, and A carries 360 N of the 600 N total. The support nearer the load takes the larger share. That is the physical direction, and it confirms the arithmetic.

The distractors.

- **300 N** is $600/2$: the total halved. That is correct only when the load is at the centre, where the two supports are symmetric. Here the load is 2.0 m from A and 4.0 m from B, so the arrangement is not symmetric and an even split is wrong.
- **240 N** is the reaction at B, and it is also the beam's own weight halved. It is a real number in this problem, which is exactly why it is offered: a candidate who solves for the wrong support, or who forgets the load entirely, lands on it.
- **480 N** comes from $(240 \times 3.0 + 360 \times 6.0)/6.0$. It takes the load's moment arm about B to be the full length of the beam, which would put the load at A itself.
- **180 N** comes from using 3.0 m for the load as well as for the beam: $(240 \times 3.0 + 360 \times 3.0)/6.0 = 1800/6.0 = 180$. It treats the load as though it also sat at the centre.

The trap. Reading the load's distance from the end it is measured from, instead of from the pivot you chose. The stem deliberately measures the load from A while the natural pivot is B; the arm you need is the one you did not write down.

Relevant topics: the principle of moments; the centre of gravity of a uniform body; choosing a pivot; the two independent checks (forces sum, moments about the other end).

The trap. Using the load's distance from the end it was measured from as its moment arm. The arm is the perpendicular distance from the pivot you chose, which here is 4.0 m and not 2.0 m.

Q2 · Dimensional analysis: the vibration frequency of a liquid droplet — **Answer D**

What is being tested. Turning a physical statement into three simultaneous equations and solving them. No formula for a droplet is needed or expected: everything is in the question.

Step 1 — write down the dimensions. Read them off the units rather than trying to recall them:

$$\begin{aligned} [f] &= T^{-1} \\ [\gamma] &= N \, m^{-1} = M \, T^{-2} \\ [\rho] &= M \, L^{-3} \\ [r] &= L \end{aligned}$$

Surface tension is the only awkward one. A newton is $kg \, m \, s^{-2}$, so a newton per metre is $kg \, m \, s^{-2} / m = kg \, s^{-2}$: mass to the first power, time to the minus two, and no length at all.

Step 2 — propose the form and match. Assume $f = k \gamma^a \rho^b r^c$, with k a dimensionless number that dimensional analysis is allowed not to find. Match each base dimension in turn:

$$\begin{array}{lcl} \text{mass:} & a + b = 0 \\ \text{length:} & -3b + c = 0 \\ \text{time:} & -2a = -1 \end{array}$$

Step 3 — the time equation alone fixes a. $-2a = -1$ gives $a = 1/2$. That is the whole reason a square root appears, and it is forced: only surface tension carries time, and it carries it as T^{-2} . If the answer had no square root, it could not be a frequency at all.

Step 4 — mass, then length. From $a + b = 0$ we get $b = -1/2$, so the density sits underneath the root. Then $-3b + c = 0$ gives $c = 3b = -3/2$: the radius appears as $r^{-3/2}$, which is the strongest power of the three and the one that makes small droplets buzz.

Step 5 — assemble and check.

$$\begin{aligned} f &= k \gamma^{1/2} \rho^{-1/2} r^{-3/2} \\ &= k \sqrt{\gamma / (\rho r^3)} \end{aligned}$$

Put the dimensions back in to be sure: $\gamma / (\rho r^3)$ has units $(\text{N/m}) / (\text{kg/m}^3 \times \text{m}^3) = \text{N m}^{-1} \text{kg}^{-1}$, and since $\text{N} = \text{kg m s}^{-2}$ that is s^{-2} . Its square root is s^{-1} , a frequency. So **Answer: D**.

The distractors.

- $\sqrt{\gamma/(\rho r)}$ is right on mass and time and takes $c = -1/2$. It is the most tempting wrong answer because it has the right shape — a root, density underneath, radius underneath. It is wrong only in the exponent on r , which is the part of the working that is easiest to rush.
- $\sqrt{\gamma r^3/\rho}$ takes $c = +3/2$, the sign of the length exponent flipped. It would make bigger droplets vibrate faster, which is the wrong way round.
- $\sqrt{\gamma \rho / r^3}$ puts the density on top, i.e. $b = +1/2$. That contradicts the mass equation $a + b = 0$, which is not $a - b = 0$.
- $\gamma/(\rho r^3)$ has all three exponents right and no root. Its units are s^{-2} , so it is a frequency squared. It can be rejected by units alone, without doing any of the exponent algebra — which is worth noticing, because it is the cheapest possible check on any answer of this kind.

The trap. Hunting for a remembered formula for a vibrating droplet. There is none worth remembering, and there does not need to be: the three exponents are forced by the dimensions, and the only freedom left is the dimensionless constant in front, which is exactly the part dimensional analysis is not able to supply.

Relevant topics: base and derived units; the dimensions of mechanical quantities; simultaneous equations in three unknowns; the limits of dimensional analysis; why small droplets ring at high frequency.

The trap. Searching for a remembered droplet formula instead of solving the three exponent equations. The time dimension appears only in the surface tension, so it alone forces the $1/2$ power and hence the root.

Q3 · Friction against a vertical wall: the smallest push that holds a block up — **Answer B**

What is being tested. Whether you see that the push and the weight act in perpendicular directions, and are connected only through friction. Candidates who treat the push as though it were holding the

block up directly get an answer that is out by a factor of $1/\mu$.

Step 1 — resolve horizontally. The only horizontal forces are the push F and the wall's normal reaction N . There is no acceleration, so they are equal:

$$N = F$$

Step 2 — resolve vertically. The block's weight is $mg = 2.0 \times 10 = 20 \text{ N}$ downwards. It is on the point of sliding *down*, so friction acts upwards and must carry the whole weight:

$$\text{friction} = mg = 20 \text{ N}$$

Step 3 — bring in the friction law. "On the point of sliding" means the friction is at its limiting value, and limiting friction is set by the normal reaction:

$$\text{friction} = \mu N = 0.40 N$$

Step 4 — eliminate N . Substituting $N = F$ from step 1:

$$\begin{aligned} 0.40 F &= 20 \\ F &= 20/0.40 = 50 \text{ N} \end{aligned}$$

So **Answer: B.**

Step 5 — check the direction and the size. The coefficient is less than one, so the push must exceed the weight it is supporting: 50 N is larger than 20 N , as it must be. If the wall were rougher, μ would be larger and the required push would fall — push less hard and the block still holds. That is the right way round.

The distractors.

- **8.0 N** is $\mu \times mg = 0.40 \times 20$. It has the right ingredients and the wrong operation: it multiplies where the balance requires a division. It is the classic inversion, and it is the wrong answer offered here that looks most like working.
- **20 N** is the weight itself. A candidate who forgets that friction is limited — and thinks the push must simply equal the weight — lands here. It also happens to be the normal reaction you would get if the push were 20 N , which makes it look internally consistent.
- **80 N** is $mg/\mu^2 = 20/0.25$. It divides by the coefficient twice, which no step of the reasoning does.
- **5.0 N** is $\mu \times mg / 4$, or equivalently 20×0.25 . It treats the coefficient as applying to the mass in kilograms rather than to a force, and gets a push smaller than the weight it is meant to support — impossible with a coefficient below one.

The trap. The instinct that a bigger push means more friction, so the smallest push is the smallest friction — and therefore a small number. The truth is the other way round: friction is limited by the normal reaction, so a smaller push means less friction available, and below 50 N the block slides.

Relevant topics: equilibrium under perpendicular forces; the normal reaction and what sets it; limiting friction; why μ less than one means the holding force exceeds the load.

The trap. Reading the push as directly supporting the weight. It sets the normal reaction, and the normal reaction sets the friction, so the required push is the weight divided by the coefficient and is larger than the weight.

Q4 · A balanced bridge: the fourth resistance — **Answer A**

What is being tested. Whether you can turn "no current in the detector" into a statement about ratios. The bridge is a potential divider used twice, and balance is the two dividers agreeing.

Step 1 — what does balance mean? If no current flows through G , then the two junctions it connects are at the same potential. There is a second, equally useful reading: the same current flows through the two resistances of the left arm, and the same current through the two of the right arm.

Step 2 — write each arm as a potential divider. Let the supply be V . The left junction sits at $V \times R_2/(R_1 + R_2)$ and the right junction at $V \times R_4/(R_3 + R_4)$. Balance says these are equal:

$$R_2/(R_1 + R_2) = R_4/(R_3 + R_4)$$

Step 3 — simplify. Cross-multiplying and cancelling the common $R_2 R_4$ term leaves the form worth remembering:

$$R_1/R_2 = R_3/R_4$$

i.e. the ratio of the left arm equals the ratio of the right arm

Step 4 — put the numbers in. With $R_1 = 3.0 \text{ k}$, $R_2 = 3.0 \text{ k}$, $R_3 = 6.0 \text{ k}$ and $R_4 = R$:

$$3.0/3.0 = 6.0/R$$

$$1 = 6.0/R$$

$$R = 6.0 \text{ k}\Omega$$

So **Answer: A**.

Step 5 — check by going back to potentials. With a 12 V supply, the left junction is at $12 \times 3.0/6.0 = 6.0 \text{ V}$, and the right junction is at $12 \times 6.0/12 = 6.0 \text{ V}$. The two agree, so no current flows through the detector. The check is worth doing because it uses the original condition rather than the simplified ratio, so it cannot inherit a mistake made while simplifying.

The distractors.

· **2.0 k Ω** is $6.0 \times 3.0/9.0$ — the ratio applied upside down, giving the reciprocal-type answer. Since the left arm has a ratio of 1, the right arm must also have a ratio of 1, so R must equal 6.0 k Ω and not something smaller.

· **18 k Ω** is $3.0 + 6.0 + 3.0 + \dots$, or 6.0×3.0 : it forms a sum or a bare product where the balance condition requires a ratio.

· **4.5 k Ω** is $3.0 \times 3.0/2$. It comes from halving a product, which is what happens if you try to use $R_1 R_2 = R_3 R_4$ — a condition that is not the balance condition.

· **3.0 k Ω** assumes all four arms are equal. That is true only for a symmetric bridge, and this one is not: the right arm already contains 6.0 k Ω .

The trap. Reaching for "the products are equal" because it sounds like the balance condition. The balance condition is about ratios. A quick dimensional sanity check helps: every term of the correct relation is a pure number, while $R_1 R_2 = R_3 R_4$ compares resistances squared, and nothing in the physics squares anything.

Relevant topics: the potential divider; the balanced Wheatstone bridge; the condition for no current in a detector; why a bridge measures a ratio and not an absolute value.

The trap. Using $R_1 R_2 = R_3 R_4$ because it looks like the balance condition. Balance is about ratios, and it is the ratio of the left arm that must equal the ratio of the right arm.

Q5 · Estimation: the power needed to climb a 10% hill on a bicycle — **Answer E**

What is being tested. Whether you can convert a gradient into a vertical speed. Everything after that is one multiplication. This is the kind of question the real paper uses to see whether a candidate is comfortable with ratios and units rather than with formulas.

Step 1 — what does a gradient of 10% mean? A gradient is a ratio of two lengths, not an angle and not a force. Ten per cent means the road rises 0.10 m for every 1.0 m travelled *along* the road. The horizontal distance is not the useful quantity here; the distance along the road is what the speed refers to.

Step 2 — turn it into a vertical speed. The cyclist covers 5.0 m of road each second. Along that road the rise is one tenth of the distance, so:

$$\text{vertical speed} = 0.10 \times 5.0 = 0.50 \text{ m/s}$$

This is the step the whole question turns on. The cyclist is rising half a metre every second, not five metres.

Step 3 — the power against gravity. Raising a mass at a steady speed requires force equal to its weight, and power is force times speed:

$$\begin{aligned} P &= m g v_{\text{vertical}} \\ &= 80 \times 10 \times 0.50 \\ &= 400 \text{ W} \end{aligned}$$

So **Answer: E.**

Step 4 — is the size right? A racing cyclist can hold a few hundred watts for an hour, and a club cyclist rather less; professional riders reach about 400 W for a sustained effort. So 400 W is a plausible figure for a hill climb, and it is the kind of cross-check that catches a factor-of-ten slip. The same computation with the full 5.0 m/s as the vertical speed would give 4000 W , which no human can deliver for more than a second or two — and that is exactly how you know it is wrong.

Step 5 — what has been left out, and why that is allowed. Rolling resistance and air resistance also take power, so the cyclist's total output is more than 400 W . The question asks for the power delivered *against gravity*, which is the part the height gain demands. Being clear about which part is being asked for is part of the skill: the estimate is deliberately of the one term you can compute from the data given.

The distractors.

- **40 W** is $80 \times 10 \times 0.05$: the gradient applied twice, or a factor of ten lost. A cyclist climbing at 400 W is a familiar figure, so 40 W should look too small for a hill.
- **2000 W** is $80 \times 5.0^2/1$ in structure — the speed squared rather than the height gained. That is the kinetic-energy form, and it has nothing to do with climbing at a steady speed, where kinetic energy does not change.
- **200 W** is $80 \times 10 \times 0.25$: the vertical speed halved. It comes from treating the 10% gradient as giving a rise of a quarter of a metre per second, which is a factor of two adrift.
- **4000 W** is $80 \times 10 \times 5.0$: the full speed used as the vertical speed, which is what a 100% gradient would give. It is the answer to a different question — climbing a wall at 5.0 m/s.

The trap. Multiplying by the gradient the wrong way, or not at all. The phrase "10%" invites you to do something with 0.10; the question is whether you multiply the speed by it or divide by it. Multiplying is right, because a smaller gradient must mean less power, and multiplying by a number below one makes the answer smaller.

Relevant topics: gradients as ratios; power as force times speed; work done against gravity; one-significant-figure estimation and the sanity check against a familiar figure.

The trap. Using the road speed as the vertical speed. A 10% gradient converts 5.0 m/s along the road into only 0.50 m/s upwards, so the power is a factor of ten smaller than the naive answer.

Q6 · Velocity-time graph: the distance travelled is the area under it — **Answer C**

What is being tested. Whether you know that the area under a velocity-time graph is a distance, and whether you can cut a compound shape into pieces whose areas you can take without a calculator.

Step 1 — what does the graph give? The vertical axis is velocity, the horizontal axis is time. A small rectangle of width dt and height v has area $v dt$, and $v dt$ is the distance covered in that instant. Adding all the slivers up gives the total distance as the whole area under the graph. The units confirm it: metres per second multiplied by seconds is metres.

Step 2 — cut the area up. The graph has three straight sections, so the area is three pieces, none of which needs any calculus:

0.0 s to 2.0 s: rises from 0 to 12 m/s, a triangle
 2.0 s to 5.0 s: flat at 12 m/s, a rectangle
 5.0 s to 8.0 s: falls from 12 m/s to 0, a triangle

Step 3 — take each piece.

triangle: $(1/2) \times 12 \times 2.0 = 12 \text{ m}$
 rectangle: $12 \times 3.0 = 36 \text{ m}$
 triangle: $(1/2) \times 12 \times 3.0 = 18 \text{ m}$
 total: $12 + 36 + 18 = 66 \text{ m}$

So **Answer: C.**

Step 4 — check the size. The peak speed is 12 m/s, so over 8.0 s the largest possible distance is $12 \times 8.0 = 96 \text{ m}$, and that would need the car to be at 12 m/s the whole time. It is not: it spends the first 2.0

s and the last 3.0 s below the peak. So the answer must be comfortably less than 96 m, and 66 m is. It must also be more than the rectangle alone, 36 m, because the two ramps add something. Both bounds hold.

The distractors.

- **96 m** is 12×8.0 : the peak speed treated as a constant. It is the answer you get by reading one number off the graph and doing one multiplication, which is exactly why the graph has ramps at both ends.
- **54 m** is $36 + 18$: the rectangle and the final ramp, with the initial ramp left out. It is a natural slip when the graph is read from left to right and the first section is mentally filed as "the bit before it starts".
- **12 m** is the peak velocity, not a distance. Mixing up what the ordinate means with what the area means is the single most common error on this kind of question.
- **84 m** is $96 - 12$: the full rectangle with one ramp removed but not the other. It looks principled — the candidate knows a triangle has to come off — but only one of the two does.

The trap. Treating the area under a straight section as though it were the triangle formula everywhere, or the rectangle formula everywhere. The middle section is flat, so it is a rectangle and there is no halving; the outer two start or finish at zero, so they are triangles and there is. Reading where each starts and stops is the real work.

Relevant topics: velocity-time graphs; the area under a graph as a physical quantity; the equations for the area of a triangle and a rectangle; unit checking as a way of confirming which quantity a graph area gives.

The trap. Reading the peak velocity as the distance, or treating the flat middle section as a triangle. The area is the answer, and the three sections need two different area formulas.

Q7 · Elastic collision: the speed of the heavier target — **Answer A**

What is being tested. Whether you know both conditions an elastic collision satisfies, and whether you use the elastic condition in the form that keeps the algebra linear.

Step 1 — momentum. Take the direction of travel as positive. Before the collision the momentum is $m u$; afterwards it is $m v_1 + 3m v_2$, where v_1 is the final velocity of the light particle (negative if it rebounds) and v_2 that of the heavy one:

$$m u = m v_1 + 3m v_2$$

$$u = v_1 + 3v_2$$

Step 2 — the elastic condition. Kinetic energy is conserved, and it is a standard result that for a head-on elastic collision this is equivalent to the relative speed of approach equalling the relative speed of separation. Before, the approach speed is u (the heavy particle is at rest); after, the separation speed is $v_2 - v_1$:

$$u = v_2 - v_1$$

Using this form rather than writing out the kinetic energy is the point of the question. Written with squares, the two equations give a quadratic and one root has to be discarded; written this way, the

substitution is one line.

Step 3 — eliminate v_1 . From the elastic condition, $v_1 = v_2 - u$. Substituting:

$$\begin{aligned} u &= (v_2 - u) + 3v_2 \\ u &= 4v_2 - u \\ 2u &= 4v_2 \\ v_2 &= u/2 \end{aligned}$$

So **Answer: A.**

Step 4 — check by conserving energy. With $v_2 = u/2$ the momentum equation gives $u = v_1 + 3u/2$, so $v_1 = -u/2$: the light particle rebounds at half its original speed. Now the energies. Before: $(1/2) m u^2$. After:

$$\begin{aligned} &(1/2) m (-u/2)^2 + (1/2) (3m) (u/2)^2 \\ &= (1/8) m u^2 + (3/8) m u^2 \\ &= (1/2) m u^2 \end{aligned}$$

The energies agree, so the answer is consistent with both conditions. The check is worth doing because it uses the energy statement directly, and so cannot inherit a mistake made in the relative-speed form.

Step 5 — does the direction make sense? A light particle hitting a heavy one head-on must bounce back, and $v_1 = -u/2$ is a rebound. A ball thrown at a wall does exactly this, and the wall is the limit of an infinitely heavy target. The heavy particle moves forwards slowly, at less than the incident speed, which is also what the limit requires.

The distractors.

- **$u/3$** comes from assuming the incident particle stops dead: $3m v_2 = m u$ gives $v_2 = u/3$. Momentum is conserved, but most of the kinetic energy has vanished, which an elastic collision does not allow.
- **$u/4$** is the perfectly inelastic result: the two move off together at $m u / (4m) = u/4$. It conserves momentum and is the smallest possible final speed, so it is the wrong end of the range.
- **u** is what an elastic collision between two *equal* masses gives: the incident particle stops and the target moves off at the full speed. It is correct for that problem, which is why it is worth having in mind — and it is wrong here only because the masses are unequal.
- **$3u/4$** splits the speed in the ratio of the masses, giving the heavy particle three quarters of u . Splitting in proportion to mass has no basis in either conservation law; the momentum equation divides by the total mass only in the perfectly inelastic case.

The trap. Using conservation of momentum alone. One equation cannot fix two unknown velocities, so a candidate who stops there is forced to guess — and every guess is one of the four wrong options. The elastic condition is not optional.

Relevant topics: conservation of momentum in one dimension; elastic and inelastic collisions; the relative-speed condition; kinetic energy as a check on a collision calculation.

The trap. Stopping after conservation of momentum. Two unknown final velocities need two equations, and the second one is the elastic condition written as approach equals separation.

Q8 · A string fixed at both ends: the third harmonic and its frequency — **Answer B**

What is being tested. Whether you count half wavelengths or full ones. Everything else in the question is a single division, so the whole mark turns on that count.

Step 1 — what the fixed ends force. A fixed end cannot move, so it must be a node. A node is a point of zero displacement, and successive nodes are half a wavelength apart. So the string's length must contain a whole number of half wavelengths, never a whole number of full ones:

$$L = n \times \lambda/2, \quad n = 1, 2, 3, \dots$$

Step 2 — read the pattern. The drawing shows three half wavelengths along the string: the string crosses the axis twice in between the ends, so there are four nodes in total (two ends and two in between) and three loops. Three loops means $n = 3$, which is the third harmonic.

$$L = 3 \lambda/2$$

$$\lambda = 2L/3 = 2 \times 0.60/3 = 0.40 \text{ m}$$

Step 3 — get the frequency. The wave speed on the string is fixed by its tension and mass per unit length, and does not change when the string is plucked differently. So the same 120 m/s applies:

$$v = f \lambda$$

$$f = v/\lambda = 120/0.40 = 300 \text{ Hz}$$

So **Answer: B.**

Step 4 — check against the fundamental. The fundamental is the case $n = 1$: $\lambda = 2L = 1.2 \text{ m}$, so $f = 120/1.2 = 100 \text{ Hz}$. The third harmonic should be exactly three times that, and $3 \times 100 = 300 \text{ Hz}$. The two routes agree, which is a strong check because they use different wavelengths.

Step 5 — check the direction. A higher harmonic means more loops, a shorter wavelength, and therefore a higher frequency. Going from 100 Hz to 300 Hz as the pattern goes from one loop to three is the right direction, and the ratio 3 matches the ratio of the loop counts.

The distractors.

- **100 Hz** is the fundamental. It is what you get by taking the wavelength to be $2L$, which is the $n = 1$ pattern rather than the one drawn. Since the drawing plainly has three loops, the harmonic number has been dropped rather than misread.
- **200 Hz** is the second harmonic, $\lambda = L = 0.60 \text{ m}$. It is one loop short of the pattern drawn, so it is the answer to a question about the graph one step to the left.
- **600 Hz** is $120/(0.60/3) = 120/0.20$: the wavelength taken to be $L/3$, which means three *full* wavelengths in the string. That is the mistake this question exists to catch. Three full wavelengths would need six nodes, and the drawing has four.
- **900 Hz** is 3×300 : the harmonic number applied twice, once in the wavelength and once again at the end. Multiplying by three twice is a common slip when the working has the number three in it in two different roles.

The trap. Reading the harmonic number as the number of full wavelengths in the string. It is the number of half wavelengths. The drawing is the guard against this: four nodes on the string is three half wavelengths, and three half wavelengths is $3 \lambda/2$.

Relevant topics: stationary waves on a string; nodes and antinodes; the harmonic series; $v = f \lambda$; why the wave speed does not change with the harmonic.

The trap. Counting full wavelengths in the string instead of half wavelengths. Three loops is three half wavelengths, so the wavelength is $2L/3$ and not $L/3$ -- a factor of two, and 600 Hz instead of 300 Hz.

Q9 · Apparent depth: how far below the surface a submerged object seems to be — **Answer E**

What is being tested. A single relation, applied the right way round. The physics fixes the direction, so the question is really about whether you divide or multiply — and one of those can be ruled out before any arithmetic.

Step 1 — what the light does. Light travelling from water into air speeds up, and when a wave speeds up as it crosses a boundary it bends away from the normal. The ray that reaches the observer's eye therefore appears to have come from a point higher up than the stone itself. The stone looks shallower than it is. This is worth fixing first, because it rules out half the options immediately.

Step 2 — the relation. For an observer looking straight down, the geometry of the ray gives a particularly clean result:

$$n = \text{real depth} / \text{apparent depth}$$

$$\text{so apparent depth} = \text{real depth} / n$$

The refractive index is greater than one, so dividing by it makes the apparent depth smaller than the real one, which is the direction step 1 required. The two agree.

Step 3 — put the numbers in.

$$\text{apparent depth} = 2.4 / 1.5 = 1.6 \text{ m}$$

So **Answer: E**.

Step 4 — check the factor. The index is 1.5, so the stone should look two thirds as far away as it is: $2.4 \times (2/3) = 1.6$. Multiplying by two thirds and dividing by 1.5 are the same operation, so the two routes agree. The answer also sits sensibly between the real depth and the surface, which is where an apparent depth always is.

The distractors.

- **3.6 m** is 2.4×1.5 : the index used as a multiplier. It is the most tempting wrong answer because the arithmetic is identical in shape, but it makes the pool look *deeper* than it is, which refraction cannot do. This is the distractor the direction check in step 1 eliminates.
- **2.4 m** is the real depth, unchanged. It is what you get if the refraction at the surface is forgotten altogether — the answer to a question about a stone in a vacuum.
- **0.80 m** is $2.4/3.0$: the index applied twice, as though the ray were bent at two surfaces each contributing a full factor. There is only one surface here.
- **1.2 m** is $2.4/2$: the depth simply halved, which would be right for an index of exactly 2 and not for 1.5.

The trap. Remembering that "something is divided by n" without remembering which thing. The safe route is to fix the direction from the physics first — a submerged object always looks shallower — and

then choose the operation that produces a smaller number.

Relevant topics: refraction at a plane surface; refractive index; why an object in water looks raised; near-normal viewing and the small-angle condition that makes the relation exact.

The trap. Multiplying by the refractive index. Dividing is right, because a submerged object always looks shallower, and the direction can be fixed from the physics before any arithmetic is done.

Q10 · A loaded potential divider: the voltage actually delivered — **Answer D**

What is being tested. Whether you notice that the load is part of the circuit, not an observer of it. A divider's output voltage is only the simple ratio while nothing is drawing current from it.

Step 1 — what is in parallel with what. The load is connected across the 3.0 kΩ resistor, so the two sit between the same two points and must be combined in parallel. That combination is the new lower arm of the divider:

$$3.0\text{ k} \parallel 3.0\text{ k} = (3.0 \times 3.0)/(3.0 + 3.0) = 9.0/6.0 = 1.5\text{ k}\Omega$$

Two equal resistors in parallel give half of one, so 1.5 kΩ can be written down without any working at all.

Step 2 — the total resistance. The upper resistor and the new lower arm are in series, since all the current from the supply passes through both:

$$R_{\text{total}} = 6.0 + 1.5 = 7.5\text{ k}\Omega$$

Step 3 — the current.

$$I = V/R = 12/7.5\text{ k}\Omega = 1.6\text{ mA}$$

Step 4 — the voltage across the load. All 1.6 mA passes through the 1.5 kΩ combination, and the load is part of that combination, so the voltage across the load is the voltage across the pair:

$$V = I R = 1.6\text{ mA} \times 1.5\text{ k}\Omega = 2.4\text{ V}$$

So **Answer: D**.

Step 5 — the check that costs nothing. Without the load, the divider would give $12 \times 3.0/9.0 = 4.0\text{ V}$. Connecting a load can only ever pull the output down, because it draws current and adds a parallel path. So the answer must be below 4.0 V, and 2.4 V is. It must also be above zero, and it is. An answer of 4.8 V or 6.0 V would be eliminated by this reasoning alone, without any calculation.

Step 6 — does the load take a sensible share? The load's 3.0 kΩ equals the resistor it sits across, so it takes half the current reaching that junction. The pair's 1.5 kΩ against the 6.0 kΩ above means the lower arm gets only one fifth of the supply: $12/5 = 2.4\text{ V}$. That agrees, and it is a quicker route once the parallel combination is recognised.

The distractors.

· **4.0 V** is the unloaded divider, $12 \times 3.0/9.0$. It is the single most common error with this circuit, and it is genuinely wrong: it uses 3.0 kΩ as the lower arm when the lower arm is 1.5 kΩ.

- **4.8 V** puts the load across the 6.0 kΩ resistor. That gives $6.0\text{ k} \parallel 3.0\text{ k} = 2.0\text{ k}$, a total of 5.0 kΩ, a current of 2.4 mA, and 2.4 mA across the 2.0 kΩ combination, which is 4.8 V. It is a fully worked answer to the wrong circuit.
- **3.0 V** assumes the supply divides equally between the 6.0 kΩ resistor and the load's branch. That would need the two to be equal, and 6.0 kΩ is not equal to 1.5 kΩ.
- **1.2 V** is exactly half of the correct answer: the load taking half of what actually reaches it, which double-counts the parallel split.

The trap. Treating the load as a measuring instrument that reads the divider's output without affecting it. A real voltmeter has a very high resistance precisely so that this approximation is good; a 3.0 kΩ load does not, and the whole question is about the difference.

Relevant topics: the potential divider; loading and output resistance; parallel and series combination; the direction check that loading always reduces an output voltage.

The trap. Reading the divider as $12 \times 3.0/9.0 = 4.0$ V. The load changes the lower arm to 1.5 k, and a loaded divider always delivers less than an unloaded one.

Q11 · Latent heat of vaporisation: how long a kettle takes to boil water away — **Answer C**

What is being tested. Whether you notice that the temperature is already fixed, so the energy is entirely about changing state. The two heat formulas look similar and are easy to swap.

Step 1 — what energy is needed? The water is at 100 °C and stays at 100 °C while it boils. Its temperature does not rise, so the formula $E = m c \Delta\theta$ contributes nothing: $\Delta\theta$ is zero. What is needed is the energy to pull the molecules apart and turn the liquid into vapour:

$$E = m L$$

where L is the specific latent heat of vaporisation. This is the whole of the physics, and it is why the question states the temperature explicitly — to tell you that the other formula is not in play.

Step 2 — put the numbers in.

$$\begin{aligned} E &= 0.50 \times 2.3 \times 10^6 \\ &= 1.15 \times 10^6 \text{ J} \end{aligned}$$

Step 3 — convert energy to time. Power is energy per unit time, so the time is the energy divided by the power:

$$\begin{aligned} t &= E/P \\ &= 1.15 \times 10^6 / 2.0 \times 10^3 \\ &= 575 \text{ s} \end{aligned}$$

So **Answer: C.**

Step 4 — check the size. 575 s is about nine and a half minutes to boil away half a kilogram of water. A kettle boils dry in a comparable time, so the answer sits where it should. This is a useful check because a factor-of-ten slip here produces 57.5 s, which is far too quick to be believable.

Step 5 — notice the direction of the comparison. Vaporising water takes much more energy per kilogram than heating it through 100 degrees: 2.3×10^6 against 4.2×10^5 , a factor of about five and a half. So the boiling stage is the slow one, which is why a kettle seems to take a long time to disappear even once it is boiling hard.

The distractors.

- **1150 s** is exactly twice the correct answer: a full kilogram used instead of the 0.50 kg stated. It is the simplest possible slip and the easiest to catch, since the question gives the mass in plain sight.
- **105 s** uses the specific heat capacity with a 100 degree rise: $0.50 \times 4200 \times 100 = 2.1 \times 10^5 \text{ J}$, giving **105 s**. This is the error the question is built around, and it is a real one: it is the time to *bring* half a kilogram of cold water to the boil, not the time to boil it away.
- **57.5 s** is ten times too quick, from reading the power as 20 kW. It fails the sanity check in step 4.
- **230 s** is $1.15 \times 10^6 / 5000$: the energy divided by two and a half times the stated power, as though more than one kettle were doing the work.

The trap. Using the formula you have practised more. The specific heat capacity appears in far more questions than the latent heat, so it is the one that comes to hand — and the only defence is to read what the water is doing. Its temperature is not changing, so $\Delta\theta$ is zero and that formula has nothing to say.

Relevant topics: specific latent heat; specific heat capacity; the difference between changing temperature and changing state; power as a rate of energy transfer.

The trap. Using $m c \Delta\theta$ instead of $m L$. The water is already at 100 degrees and stays there, so the temperature rise is zero and the latent heat is the only term that matters.

Q12 · Nuclear density: why it is the same in every nucleus — **Answer A**

What is being tested. Whether you can follow a proportionality through a cube. The radius law is given, so nothing has to be remembered; the whole question is what happens when it is put into the density.

Step 1 — the mass. A nucleus is made of A nucleons, each of very nearly the same mass, and the electrons contribute almost nothing. So:

m proportional to A

Step 2 — the volume. A nucleus is very nearly spherical, so its volume is proportional to the cube of its radius:

V proportional to r^3
 $r = r_0 A^{1/3}$, so $r^3 = r_0^3 A$
 V proportional to $r_0^3 A$

The cube root in the radius law exists precisely so that the volume comes out proportional to A , and not to some awkward power of it. That is worth noticing: the law is written that way because that is the experimental fact, and the convenient algebra is a consequence of the physics rather than the other way round.

Step 3 — divide.

$$\begin{aligned}\rho &= m/V \\ &\propto A/(r_0^3 A) \\ &= 1/r_0^3\end{aligned}$$

The A cancels. The density depends only on the constant r_0 and the nucleon mass, and not on how big the nucleus is. So **Answer: A** — it is unchanged.

Step 4 — check with the numbers given. Going from $A = 27$ to $A = 216$ multiplies A by $216/27 = 8$. The mass goes up by 8 and the volume also goes up by 8, because the volume is proportional to A . So the ratio is $8/8 = 1$. The density does not move.

Step 5 — why this is a striking result. Nuclear matter has the same density in a hydrogen nucleus as in a lead nucleus, about $2 \times 10^{17} \text{ kg/m}^3$. It is a liquid-like incompressibility: adding more nucleons makes the nucleus bigger rather than denser. That is a real physical fact about the strong force, and this question is a way of seeing it fall out of the radius law.

The distractors.

- **Increases by a factor of 8** scales the density with A . It uses the mass growth and forgets the volume growth, which is the whole content of the question.
- **Decreases by a factor of 8** is the inverse: it uses the volume growth and forgets the mass. It would be right if the volume grew like A^2 , and the radius law is the statement that it does not.
- **Increases by a factor of 2** applies the cube root of 8 to the density. The cube root of 8 is 2, but it is the *radius* that doubles, not the density. The density depends on the cube of the radius, and the cube of 2 is 8, which is exactly what cancels the mass.
- **Decreases by a factor of 2** applies the radius change to the density with the sign reversed. The density has no dependence on the radius left in it at all, so neither a factor of 2 nor its reciprocal can appear.

The trap. Treating the density as though it followed the radius. The radius grows as $A^{(1/3)}$ and the density does not grow at all, because the mass grows in step with the volume. Whenever a quantity cancels, it is worth saying out loud which two things cancelled — here, the A in the mass against the A in the volume.

Relevant topics: the nuclear radius law; proportional reasoning through a cube; density as mass over volume; why all nuclei have the same density.

The trap. Scaling the density with A . The mass goes as A and the volume goes as A too, because the radius law carries a cube root, so the two cancel and the density is independent of the size of the nucleus.

Q13 · Scaling: how the total surface area changes when a solid is cut up — **Answer E**

What is being tested. Whether you can hold two scalings in mind at once — the pieces get more numerous, and each piece gets smaller — and combine them. Doing only one of the two is what the wrong options are.

Step 1 — how much smaller is each piece? One cube becomes 1000 equal cubes, so each has $1/1000$ of the volume. Volume scales as the cube of a length, so a linear dimension must be divided by the cube root of 1000:

$$(\text{side of a small cube})/(\text{side of the original}) = 1/10$$

Equivalently, $10^3 = 1000$, which can be seen directly.

Step 2 — how much smaller is each piece's area? Area scales as the square of a linear dimension, so a reduction by 10 in length is a reduction by $10^2 = 100$ in area:

$$(\text{area of one small cube})/(\text{area of the original}) = 1/100$$

Step 3 — combine the two. There are 1000 pieces, each contributing $1/100$ of the original area:

$$\begin{aligned} \text{total area} &= 1000 \times (1/100) \times A_{\text{original}} \\ &= 10 \times A_{\text{original}} \end{aligned}$$

So **Answer: E** — a factor of 10.

Step 4 — check that the answer sits where it should. The total must be larger than the original, because cutting creates new surfaces. It must be smaller than 1000 times the original, because each piece is smaller than the original. A factor of 10 is between 1 and 1000, so it passes both bounds. Both wrong bounds are among the options, which is a reminder that checking a bound is often enough to eliminate two answers at once.

Step 5 — the general rule, which is worth having. Cutting a solid into n equal pieces multiplies the total surface area by $n^{(1/3)}$. Here $1000^{(1/3)} = 10$. This is why a powder reacts so much faster than a lump, and why dust can be explosive when the same material in bulk is not: the chemistry is identical and the available surface is not.

The distractors.

- **1000** is the number of pieces. It ignores that each piece is 100 times smaller in area, which is a very large correction — it is not a small effect that can be neglected.
- **100** is the per-piece area factor, applied in the wrong direction and without the count. One small cube has $1/100$ of the original area, not 100 times it.
- **100000** is 1000×100 : the number of pieces times the area factor the wrong way up. It multiplies where one of the two factors must divide.
- **32** is $1000^{(1/2)}$, rounded. The square root belongs to a comparison of lengths if the shape is scaled in one dimension only; here the pieces are three-dimensional and the cube root is the right one.

The trap. Doing one scaling and stopping. Both "there are 1000 pieces" and "each piece is 100 times smaller in area" are true, and the answer is neither of them: it is what happens when the two are combined. The general form $n^{(1/3)}$ is a useful thing to carry, because it makes the combination automatic.

Relevant topics: scaling of area and volume with length; the square-cube law; surface-area-to-volume ratio and why it matters for reaction and cooling rates.

The trap. Doing only one of the two scalings. There are 1000 pieces AND each is 100 times smaller in area, so the total grows by $1000/100 = 10$, which is neither 1000 nor 100.

Q14 · A particle whose deceleration is proportional to its speed: the distance to rest — **Answer B**

What is being tested. Whether you can change the independent variable when the question asks for a distance but the law is about speed. Time is not mentioned anywhere in the answer, which is a strong hint that time should be eliminated.

Step 1 — why the given form is awkward. The law $a = -kv$ relates acceleration to speed. In its natural form $dv/dt = -kv$ it tells you how the speed falls with *time*, and solving it gives $v = u e^{(-kt)}$. That is true and useful, but it is not what is asked for, and turning it into a distance needs a further integration. There is a shorter route.

Step 2 — change the variable. Acceleration can be written in two ways, and the second is the one for a question about distance:

$$a = dv/dt = v \, dv/dx$$

The identity comes from the chain rule, $dv/dt = (dv/dx)(dx/dt)$, and dx/dt is v . Substituting the given law:

$$v \, dv/dx = -k v$$

Step 3 — divide by v. For v not zero this gives:

$$dv/dx = -k$$

Remarkably, the speed has dropped out. The statement is now that the speed falls at a constant rate *with distance* — a very different statement from a constant deceleration in time, and the reason the answer is not the familiar $u^2/(2a)$.

Step 4 — integrate. Separating and integrating between the starting speed u and rest:

$$\begin{aligned} x &= \text{integral of } dx = \text{integral from } u \text{ to } 0 \text{ of } (-dv/k) \\ &= -[v/k] \text{ from } u \text{ to } 0 \\ &= (u - 0)/k \\ &= u/k \end{aligned}$$

So **Answer: B**.

Step 5 — check the dimensions and the behaviour. The constant k has units of s^{-1} , because $a = -kv$ has $m \, s^{-2}$ on the left and k times $m \, s^{-1}$ on the right. So u/k has units $(m \, s^{-1})/(s^{-1}) = m$, a distance. Good. And the behaviour is right in both directions: start faster and you go further; make k larger and the motion is damped harder, so you go less far. Both are what the physics requires.

Step 6 — a second route, for confirmation. From $v = u e^{(-kt)}$, the distance is the integral of the speed from $t = 0$ to infinity:

$$x = \text{integral of } u e^{(-kt)} \, dt = u/k$$

The two routes agree, and they use different mathematics — one integrates in space, the other in time. That is the strongest kind of check available here.

The distractors.

- **uk** is the reciprocal of the right answer, from multiplying by **k** where the working divides. A quick units check kills it: **u k** has units $\text{m s}^{-1} \times \text{s}^{-1} = \text{m s}^{-2}$, which is an acceleration and not a distance.
- **u/(2k)** is half the correct answer. It comes from reaching for the constant-deceleration formula $x = u^2/(2a)$ and putting $a = ku$, which is only the initial value of the deceleration. The deceleration is not constant, so that formula does not apply.
- **2u/k** is twice the correct answer. It treats the distance to rest as twice the distance covered while the speed halves — plausible-looking, but the speed does not fall linearly, so halving the speed does not take half the distance.
- **u/(k ln 2)** uses the *time* to halve the speed. The speed halves after $t = \ln 2/k$, and dividing **u** by that time gives a quantity with the units of an acceleration, not a distance. It is a real quantity from this problem, applied to the wrong question.

The trap. Using the constant-acceleration formulas because they are the ones you have practised. They require the acceleration to be constant, and here it falls in step with the speed, so they are simply not applicable. The question deliberately gives a law that is not constant in time, and the identity $a = v \, dv/dx$ is the tool for that case.

Relevant topics: the two forms of acceleration; separating variables; the chain rule in kinematics; why an exponential decay takes infinite time but finite distance.

The trap. Applying the constant-acceleration formulas. The deceleration falls as the speed falls, so $u^2/(2a)$ is not available, and the identity $a = v \, dv/dx$ is the way in.

Q15 · Impulse from a force-time graph, and the speed it gives a ball — **Answer D**

What is being tested. Whether you read a force-time graph the same way as a velocity-time graph, and whether you then complete the step from momentum to speed.

Step 1 — what the area means. A force acting for a short time dt changes the momentum by $F \, dt$. Adding up all the slivers, the total change in momentum is the whole area under the force-time graph. This area has its own name, the impulse, and its units are newton seconds. Since a newton is a kg m s^{-2} , a newton second is a kg m s^{-1} — the units of momentum, as they must be.

Step 2 — take the area. The graph is a triangle rising from zero to 20 N and back to zero over 0.20 s:

$$\begin{aligned} \text{impulse} &= (1/2) \times \text{base} \times \text{height} \\ &= (1/2) \times 0.20 \times 20 \\ &= 2.0 \, \text{N s} \end{aligned}$$

Step 3 — from impulse to speed. The ball starts at rest, so the change in momentum is the final momentum itself:

$$\begin{aligned} \text{impulse} &= \Delta p = m \, \Delta v \\ 2.0 &= 2.0 \times v \\ v &= 1.0 \, \text{m/s} \end{aligned}$$

So **Answer: D.**

Step 4 — check the size. An impulse of 2.0 N s is the same as the momentum of a 2.0 kg ball moving at 1.0 m/s, which is a brisk walking pace — a firm push on a ball for a fifth of a second. A kick in a football match delivers a much larger impulse, of order 10 N s, so 2.0 N s giving 1.0 m/s is the right order for the force and time given.

Step 5 — the comparison worth carrying away. Both graph questions in this paper are read the same way. On a velocity-time graph the area is a distance; on a force-time graph the area is a momentum. In each case the vertical axis is a rate of change of the quantity the area gives: velocity is the rate of change of distance, force is the rate of change of momentum. That is not a coincidence, and it is a reliable way to remember which area gives what.

The distractors.

- **2.0 m/s** is the impulse reported as a speed. The number 2.0 appears twice in this problem — once as the impulse and once as the mass — so a candidate who stops after step 2 lands on an answer that looks internally consistent.
- **0.50 m/s** is $2.0/4.0$: the mass applied twice. It is the natural slip when both the impulse and the mass are 2.0, since it is easy to divide twice without noticing.
- **4.0 m/s** reads the peak force of 20 N as though it had acted for the whole 0.20 s. That gives an impulse of $20 \times 0.20 = 4.0 \text{ N s}$ and a speed of 2.0 m/s. It is the rectangle the triangle sits inside, and the graph's triangular shape is the whole reason the answer is not that.
- **20 m/s** is the peak force itself, read off the vertical axis. It mixes up what the axes mean — the ordinate is a force, not a speed.

The trap. Reading the graph as though it were a rectangle. The peak value is the eye-catching number, and multiplying it by the duration is one step with no halving in it. The halving is what the sloping sides demand, and the graph is drawn with slopes on purpose.

Relevant topics: impulse and momentum; the area under a force-time graph; the impulse-momentum equation; reading compound areas; the parallel between the two kinds of graph.

The trap. Treating the force-time graph as a rectangle and using the peak force for the whole duration. The triangular area is half of that, so the impulse and the speed are both half of the naive answer.

Q16 · A banked track: the angle at which no friction is needed — **Answer A**

What is being tested. Whether you can resolve the normal reaction into components and then combine the two equations in the one way that removes the unknowns. There is no friction in the problem, so the normal reaction is the only force that can point towards the centre.

Step 1 — what the normal reaction is doing. On a banked surface the normal reaction is perpendicular to the road, so it is tilted from the vertical by the banking angle θ . That tilt is the whole point: part of it holds the car up, and part of it pushes the car towards the centre of the circle. Friction is stated to be unnecessary, so these two components are the entire story.

vertical: $N \cos(\theta) = mg$
 horizontal: $N \sin(\theta) = m v^2/r$

The second equation is the centripetal requirement: the horizontal force must equal $m v^2/r$.

Step 2 — divide one by the other. This is the step the question is really testing. Dividing the horizontal equation by the vertical one cancels N on the left, and cancels m on the right:

$$\begin{aligned} N \sin(\theta) / N \cos(\theta) &= (m v^2 / r) / (m g) \\ \tan(\theta) &= v^2 / (r g) \end{aligned}$$

Two unknowns gone in one operation. Nothing else in the problem is as efficient as this, and it is worth remembering as the standard move whenever two resolved equations share a common force.

Step 3 — put the numbers in.

$$\begin{aligned} \tan(\theta) &= 15^2 / (30 \times 10) \\ &= 225 / 300 \\ &= 0.75 \end{aligned}$$

Step 4 — recognise the angle. A tangent of 0.75 is the 3-4-5 triangle: $0.75 = 3/4$, so the angle is the one whose sine is $3/5$ and whose cosine is $4/5$. That angle is 37° . So **Answer: A**.

Step 5 — check the physics. A steeper bank should allow a faster corner, so the angle should grow with speed and shrink with radius: $\tan(\theta) = v^2 / (rg)$ has v^2 on top and r underneath, which is exactly right. The mass does not appear at all, so a loaded lorry and a motorcycle need the same banking angle — a result worth noticing, and one that the division in step 2 produced automatically.

The distractors.

- 53° is the angle whose tangent is $4/3$: the ratio inverted. It corresponds to $rg/v^2 = 4/3$, which is the equation the other way up. It is a real angle in the 3-4-5 family, which is what makes it convincing.
- 45° is what you get from $\tan(\theta) = 1$, i.e. from assuming the vertical and horizontal components of the normal reaction are equal. Nothing in the problem says they are; it is an assumption, not a derivation.
- 49° treats 0.75 as a sine rather than a tangent. The vertical and horizontal components of N are its cosine and sine, and it is their *ratio* that the division produces — that is precisely why a tangent appears and not a sine.
- 27° is $\text{atan}(0.5)$: the ratio computed as $v/(rg) = 15/30$, with the square on the speed dropped. The centripetal force depends on v^2 , so the square cannot be dropped.

The trap. Writing the centripetal force as $N \sin(\theta)$ and then also using $N = mg$. That double-use of the normal reaction throws away the tilt, and it produces an angle that depends on the mass — which the correct answer does not. If your answer has a mass in it, something has gone wrong.

Relevant topics: resolving forces on an inclined surface; centripetal acceleration; the banking angle; why the mass cancels; the 3-4-5 triangle as a standard angle.

The trap. Using $N = mg$ as well as $N \sin(\theta) = mv^2/r$, which uses the normal reaction twice and throws away the tilt. Dividing the two resolved equations removes N and m together, and the mass must not appear in the answer.

What is being tested. Whether you keep the two stages of the calculation separate. The stress gives an area; the question asks for a diameter. Candidates who stop at the area, or who muddle the circle relation, land on the wrong options.

Step 1 — the force the cable must carry. The load's weight, since the cable hangs vertically:

$$W = m g = 1000 \times 10 = 1.0 \times 10^4 \text{ N}$$

Step 2 — the area the stress allows. Stress is force per unit area, so the smallest acceptable area is the load divided by the breaking stress:

$$\begin{aligned} A &= W/\text{stress} \\ &= 1.0 \times 10^4 / 1.0 \times 10^9 \\ &= 1.0 \times 10^{-5} \text{ m}^2 \end{aligned}$$

A dimensional check is worth doing here, because it costs nothing: N/Pa is $(\text{kg m s}^{-2})/(\text{kg m}^{-1} \text{s}^{-2}) = \text{m}^2$, an area. Good.

Step 3 — from area to diameter. The cross-section is a circle:

$$\begin{aligned} A &= \pi d^2/4 \\ d^2 &= 4A/\pi \\ &= 4 \times 1.0 \times 10^{-5} / 3.14 \\ &= 1.3 \times 10^{-5} \\ d &= \sqrt{1.3 \times 10^{-5}} \\ &= 3.6 \times 10^{-3} \text{ m} \\ &= 3.6 \text{ mm} \end{aligned}$$

So **Answer: C.**

Step 4 — check the size. Three and a half millimetres is a few millimetres, which is about right for a steel cable lifting a tonne: steel's breaking stress of $1.0 \times 10^9 \text{ Pa}$ means a square millimetre holds about a kilonewton, so ten square millimetres holds the ten kilonewtons of a tonne, and ten square millimetres is a circle a little over three millimetres across. The size is right, which rules out the 36 mm option immediately.

Step 5 — note the two-stage structure. The most common way to lose this question is to treat d as though it were the area. Stress times area gives a force; stress times diameter does not. Writing the units beside each line is what catches this, since an answer in metres and an answer in square metres cannot be confused once they are written down.

The distractors.

- **1.8 mm** comes from using $A = \pi d^2$, i.e. putting the diameter where the radius belongs. That makes the area four times too large for a given d , so the d that satisfies it is half the correct one. The error is entirely in the circle relation, not in the stress.
- **3.2 mm** is $\sqrt{A}$ directly: the factor of $4/\pi$ dropped. That is the same as assuming the cross-section is a square of side d , which for a circular cable it is not.
- **7.1 mm** is twice the correct value — the diameter reported as though it were the radius, or the radius doubled at the end.

· **36 mm** is ten times the correct value, from reading the breaking stress as 1.0×10^8 . It is the size of a structural cable rather than a lifting cable, and the order-of-magnitude check in step 4 is what rules it out.

The trap. Stopping after the area. The stress relation is the part that has been practised, so the calculation feels finished once A is known — but the question asked for a diameter, and one more relation with a square root in it stands between the two.

Relevant topics: stress and strain; the breaking stress; the area of a circle; dimensional checking as a way of confirming which stage you have reached.

The trap. Stopping once the area is found. Stress gives an area, and the diameter needs the circle relation on top, which contributes both a square root and a factor of four over pi.

Q18 · The highest diffraction order a grating can produce — **Answer E**

What is being tested. Whether you can convert a line density into a spacing, and then recognise that the answer is bounded by an inequality rather than given by an equation.

Step 1 — the grating spacing. "300 lines per millimetre" means the lines are one three-hundredth of a millimetre apart:

$$\begin{aligned} d &= 1 \text{ mm}/300 \\ &= 1.0 \times 10^{-3} \text{ m}/300 \\ &= 3.3 \times 10^{-6} \text{ m} \end{aligned}$$

This is the step where most of the arithmetic lives, and it is worth writing the unit out: d is a length, and it must come out of the reciprocal of a density of lines.

Step 2 — the grating equation. Maxima appear where waves from adjacent slits arrive in step:

$$d \sin(\theta) = n \lambda$$

with n a whole number. Rearranged, the order is $n = d \sin(\theta)/\lambda$.

Step 3 — the limit. This is the step the question is built around. The angle of a maximum cannot exceed 90° , so $\sin(\theta)$ can be at most 1. Substituting that largest possible value gives the largest possible order:

$$\begin{aligned} n_{\text{max}} &= d/\lambda \\ &= 3.3 \times 10^{-6}/5.0 \times 10^{-7} \\ &= 6.6 \end{aligned}$$

Step 4 — round the inequality down. The statement is not "the order is 6.6"; it is " n must be a whole number no greater than 6.6". The whole numbers satisfying that are 0, 1, 2, 3, 4, 5, 6, so the highest order is **6**. So **Answer: E**.

Step 5 — check by putting the order back in. For $n = 6$, $\sin(\theta) = 6 \times 5.0 \times 10^{-7}/3.3 \times 10^{-6} = 0.90$, which is a real angle. For $n = 7$, $\sin(\theta) = 7 \times 5.0 \times 10^{-7}/3.3 \times 10^{-6} = 1.05$, which does not exist. So 6 is the last order that fits, and 7 is one too many. This check is the honest way to settle it, and it takes one line.

The distractors.

- **7** rounds 6.6 up to the nearest whole number. Rounding to nearest is a habit from measurement, and it is wrong here: the question is about which whole numbers *satisfy an inequality*, and that always rounds down.
- **12** doubles the order to count the maxima on both sides of the centre. There are indeed maxima at both $+6$ and -6 , but the question asks for the order, which is a property of one maximum and is the same on both sides.
- **13** is $2 \times 6 + 1$: both sides counted and the central maximum added. It is the right answer to "how many maxima can be seen in total", which is a different question. This pair of options — 12 and 13 — is there to separate the order from the count.
- **1** concludes that only the central maximum exists. That would require the wavelength to exceed the grating spacing, and here 500 nm is much smaller than 3300 nm , so plenty of orders fit. Checking that comparison first is a quick way to rule this out.

The trap. Rounding 6.6 to 7. The whole question is a limiting case, and in a limiting case the inequality governs, not the arithmetic habit. Whenever the answer must be a whole number and it comes out between two, ask which side the physics allows — here, only the smaller one.

Relevant topics: the diffraction grating; the grating equation; the condition $\sin(\theta) \leq 1$; rounding an inequality; the difference between the order and the number of visible maxima.

The trap. Rounding the ratio 6.6 up to 7. The order is bounded by $\sin(\theta)$ at most 1, which is an inequality, and inequalities round DOWN -- the seventh order would need a sine of 1.05.

Q19 · A converging lens: the linear magnification — **Answer B**

What is being tested. Whether you keep the lens equation and the magnification separate. They are two different statements, and the first does not give the second.

Step 1 — the lens equation. With the usual convention that f is positive for a converging lens and the object is on the incoming side:

$$1/f = 1/u + 1/v$$

where $u = 30 \text{ cm}$ and $f = 20 \text{ cm}$. The object is outside the focal length, so the image will be real and on the far side of the lens, and v should come out positive.

Step 2 — solve for the image distance. Put the two fractions over a common denominator of 60:

$$\begin{aligned} 1/v &= 1/f - 1/u \\ &= 1/20 - 1/30 \\ &= 3/60 - 2/60 \\ &= 1/60 \\ v &= 60 \text{ cm} \end{aligned}$$

The image is 60 cm from the lens, on the far side. The sign is right: a positive image distance means a real image, which is what a converging lens gives for an object outside its focal length.

Step 3 — the magnification. Linear magnification is the ratio of the image height to the object height, and the ray geometry makes that equal to the ratio of the distances:

$$\begin{aligned} m &= v/u \\ &= 60/30 \\ &= 2 \end{aligned}$$

So **Answer: B.**

Step 4 — check against the general case. Combining the lens equation with $m = v/u$ gives the compact form:

$$m = f/(u - f) = 20/(30 - 20) = 2$$

The two routes agree, which is a worthwhile check because this form never produces v at all. The behaviour also checks out: as the object moves out towards infinity, $u - f$ grows and m falls towards zero, which is what happens when you photograph something very far away. As the object approaches the focal point, m grows without limit, which is the projector case.

Step 5 — is the size sensible? The object sits between $f = 20 \text{ cm}$ and $2f = 40 \text{ cm}$, and that is exactly the arrangement that produces a real, inverted, magnified image — the arrangement in a slide projector. So a magnification of 2 with a real image is the right qualitative answer, and it confirms the direction as well as the size.

The distractors.

- **3** is $v/f = 60/20$. It divides by the focal length rather than by the object distance. Both are lengths in the working, and mixing them up is easy when all three numbers are on the page.
- **1.5** is $u/f = 30/20$. The object distance divided by the focal length appears in the working — the object is one and a half focal lengths away — but it is a statement about where the object is, not about how big the image is.
- **0.67** is $f/v = 20/60$, the reciprocal of the ratio of image distance to focal length. It is smaller than one, which would mean a diminished image, and the arrangement here does not give one.
- **0.5** is $u/v = 30/60$: the magnification upside down. It is the single most common slip with this formula, and it is worth guarding against by noting that a magnification of 0.5 would mean the image is smaller than the object, which contradicts the ray diagram.

The trap. Stopping at v . Finding the image distance is the bulk of the arithmetic, so it feels like the end of the question — but the magnification is a second ratio, and the lens equation does not contain it. Reading the ray diagram is the guard: it plainly shows an image taller than the object.

Relevant topics: the thin lens equation; real and virtual images; linear magnification; the three standard object positions and the images they give.

The trap. Stopping at the image distance, or forming the ratio with the focal length. Magnification is v/u , and both lengths in it come from the lens equation's solution, not from the data.

Q20 · Two lamps of different power rating in series: which is brighter — **Answer D**

What is being tested. Whether you can turn a power rating into a resistance, and then notice that the answer to "which is brighter" depends on how the lamps are connected.

Step 1 — what the markings mean. "60 W at 240 V" is a statement about the lamp when 240 V is across it. From $P = V^2/R$:

$$R = V^2/P$$

$$R_{60} = 240^2/60 = 57600/60 = 960 \text{ ohm}$$

$$R_{100} = 240^2/100 = 57600/100 = 576 \text{ ohm}$$

So the *lower* rated power belongs to the *higher* resistance. That inversion is the heart of the question. A 60 W lamp has a thinner, longer filament than a 100 W lamp, and a thinner filament is a larger resistance.

Step 2 — what is common in series. In series, the same current passes through both lamps. This is the opposite of parallel, where the same voltage appears across both. So the right form of the power relation is:

$$P = I^2 R$$

with I the same for both lamps.

Step 3 — compare. Since I is common and $R_{60} > R_{100}$, the power in the 60 W lamp is the larger. So **Answer: D** — the 60 W lamp is brighter.

Step 4 — check the numbers. The resistances differ by $960 - 576 = 384 \text{ ohm}$, which is the quantity computed in the check. The total resistance is $960 + 576 = 1536 \text{ ohm}$, so the current is $240/1536 \text{ A}$, which cancels to $5/32 \text{ A}$. Because the current is common, the two powers are in the ratio of the resistances, and $960 : 576 = 5 : 3$. So the 60 W lamp takes five eighths of the total power:

$$P_{\text{total}} = V^2/R = 240^2/1536 = 37.5 \text{ W}$$

$$P_{60} = (5/8) \times 37.5 = 23 \text{ W}$$

$$P_{100} = (3/8) \times 37.5 = 14 \text{ W}$$

Both are well below their ratings, which is right: each lamp only gets part of the 240 V, so neither can reach its rated power. And the 60 W lamp takes the larger share, confirming step 3 numerically.

Step 5 — the general rule, which is what makes this memorable. In parallel, the higher-power lamp is brighter, because the voltage is common and $P = V^2/R$ rewards the smaller resistance. In series, the lower-power lamp is brighter, because the current is common and $P = I^2 R$ rewards the larger resistance. The two arrangements give opposite rankings, and the question is testing whether you know which one you are in.

The distractors.

- **The 100 W lamp because it is rated higher** is the answer for a parallel connection. It is the most common error, and it comes from carrying over a habit without checking the arrangement.
- **Neither, because the current is the same** is a genuine confusion between a shared quantity and a shared outcome. Equal current does not mean equal power, because the resistances differ. The whole calculation in step 4 shows the powers are 23.5 W and 14.1 W.
- **The 100 W lamp because its lower resistance lets more current through** has the reasoning backwards. In series there is only one path, so the current cannot be different in the two lamps. The lower resistance carries the same current and therefore dissipates *less* power.

· **Neither lights** mistakes a series connection for a fault. Two lamps in series share the supply and both glow, dimly. The total resistance of 1536 ohm draws a perfectly ordinary current from a 240 V supply.

The trap. Reasoning from the labels instead of from the circuit. "100 W" is a bigger number than "60 W", and it is very tempting to conclude that the bigger number is the brighter lamp. The label describes a different circuit — the one with 240 V across that lamp alone — and it does not survive being reconnected.

Relevant topics: power ratings and what they mean; $P = V^2/R$ and $P = I^2 R$; series and parallel connections; why the two arrangements rank lamps differently.

The trap. Assuming the higher-rated lamp is brighter. That is true in parallel, where the voltage is common. In series the current is common and the power goes as resistance, so the 60 W lamp wins.

Q21 · Capacitors in series: how the supply divides between them — **Answer A**

What is being tested. Whether you know what is common in a series circuit. For resistors in series it is the current; for capacitors in series it is the charge. Getting that one fact right makes the rest a division.

Step 1 — what is shared. Two capacitors in series are joined by a single wire between them. That wire is isolated, so no charge can accumulate on it, and the charge that leaves one plate must arrive at the other. So:

$$Q_1 = Q_2 = Q$$

This is the analogue of equal current in series resistors, and it is the fact the whole question turns on.

Step 2 — the combination. The series rule is that the reciprocals add:

$$\begin{aligned} 1/C &= 1/C_1 + 1/C_2 \\ &= 1/2.0 + 1/6.0 \\ &= 3/6.0 + 1/6.0 \\ &= 4/6.0 \\ C &= 6.0/4 = 1.5 \text{ uF} \end{aligned}$$

The combination is smaller than the smallest capacitor, which is the right direction for a series arrangement — adding components in series never increases the total.

Step 3 — the common charge. The whole supply appears across the pair, so the charge stored is:

$$Q = C V = 1.5 \text{ uF} \times 12 \text{ V} = 18 \text{ uC}$$

Step 4 — divide that charge by each capacitance. Both capacitors carry 18 μC , and each has its own voltage:

$$\begin{aligned} V_2 &= Q/C_1 = 18/2.0 = 9.0 \text{ V} \\ V_6 &= Q/C_2 = 18/6.0 = 3.0 \text{ V} \end{aligned}$$

The question asks for the 2.0 μF capacitor, so the answer is **9.0 V**. So **Answer: A**.

Step 5 — the check that settles it. The two voltages must add to the supply: $9.0 + 3.0 = 12 \text{ V}$. They do, which is a complete check on both voltages at once. A second check on the direction: the smaller

capacitance takes the larger voltage, and $2.0\ \mu\text{F}$ is smaller than $6.0\ \mu\text{F}$, so $9.0\ \text{V}$ going to the $2.0\ \mu\text{F}$ capacitor is the right way round.

Step 6 — a shortcut worth having. Since the charge is common, $V_1/V_2 = C_2/C_1$: the voltages are in the *inverse* ratio of the capacitances. Here that is $6.0/2.0 = 3$, so the voltages are in the ratio 3:1, and with $12\ \text{V}$ to share they are $9.0\ \text{V}$ and $3.0\ \text{V}$. That route avoids computing the combination or the charge at all.

The distractors.

- **$3.0\ \text{V}$** is the voltage across the $6.0\ \mu\text{F}$ capacitor — the other one. It is also what you get by taking the voltages in proportion to the capacitances, which is the parallel rule. In parallel the larger capacitor does take the larger charge, so this confusion has a real source.
- **$6.0\ \text{V}$** splits the supply equally. That would need the two capacitances to be equal, and $2.0\ \mu\text{F}$ and $6.0\ \mu\text{F}$ are not.
- **$4.0\ \text{V}$** is one third of the supply: the ratio 1:2 taken the wrong way round, or the ratio 1:3 applied to the wrong capacitor.
- **$12\ \text{V}$** gives the whole supply to one capacitor, as though the other were not there. If it were alone across the supply that is exactly what it would take, which is why the option is plausible — but then the question would not have mentioned a second capacitor.

The trap. Carrying the resistor habit across. In a series resistor circuit the current is common and the voltages go as the resistances, so the *larger* component takes the larger voltage. In a series capacitor circuit the charge is common and the voltages go as the reciprocal of the capacitance, so the *smaller* component takes the larger voltage. The two arrangements rank their components in opposite orders, and the question is testing which one you are in.

Relevant topics: capacitance and $Q = CV$; capacitors in series and in parallel; what is common in each kind of connection; the inverse ratio of voltages in series.

The trap. Applying the parallel rule to a series circuit. In series the charge is common, so the voltages go as the reciprocal of the capacitance and the smaller capacitor takes the larger share.

Q22 · Linear thermal expansion: how much longer a heated rod becomes — **Answer C**

What is being tested. A proportionality applied with the right temperature. The physics is one line; the whole question is whether the rise or the reading goes in.

Step 1 — the statement. For a solid rod, the change in length is proportional to the original length and to the change in temperature:

$$\Delta L = L_0 \alpha \Delta T$$

where α is the coefficient of linear expansion. The coefficient is a property of the material, so brass's value applies to any brass rod.

Step 2 — the temperature change. The rod starts at $20\ ^\circ\text{C}$ and ends at $120\ ^\circ\text{C}$, so:

$$\Delta T = 120 - 20 = 100\ ^\circ\text{C}$$

This is the step to slow down on. The formula needs the *change* in temperature, and the stem deliberately gives a starting temperature so that a change has to be worked out. Using 120 by mistake inflates the answer by a fifth.

Step 3 — put the numbers in.

$$\begin{aligned} \Delta L &= 2.0 \times 1.9 \times 10^{-5} \times 100 \\ &= 3.8 \times 10^{-3} \text{ m} \end{aligned}$$

The coefficient is a pure number per degree, so the degrees cancel and the answer comes out in metres, the same unit as L_0 . That is the unit check: the length unit must be whatever L_0 was in.

Step 4 — convert to a sensible unit. $3.8 \times 10^{-3} \text{ m}$ is 3.8 mm. So **Answer: C**.

Step 5 — check the size. Brass expands by about nineteen parts per million per degree. Over a hundred degrees that is about nineteen parts in ten thousand, or about two parts in a thousand. Two metres times two parts in a thousand is four millimetres. So 3.8 mm is right, and it is also the right order for something you can measure with a ruler — which is why expansion joints are needed in bridges and railway lines, and why a surveyor's tape is calibrated at a stated temperature.

Step 6 — note what the answer does not depend on. The expansion does not depend on the rod's cross-section, only on its length. A thick brass rod and a thin one of the same length expand by the same amount. That is worth noticing because it is not true of every thermal property: heat capacity depends on mass, and expansion does not.

The distractors.

- **7.6 mm** is twice the correct value. It comes from doubling the coefficient, or from treating the expansion as though it happened along two axes. Linear expansion is a one-dimensional statement, and the coefficient given is for length.
- **4.6 mm** uses 120 degrees as the change: $2.0 \times 1.9 \times 10^{-5} \times 120 = 4.6 \times 10^{-3} \text{ m}$. It is the error the starting temperature of 20 °C exists to catch.
- **1.9 mm** uses a one-metre rod: $1.0 \times 1.9 \times 10^{-5} \times 100$. The length is in plain sight in the stem, so this is a careless slip rather than a conceptual one — and it is exactly the slip that writing the formula with symbols first prevents.
- **0.38 mm** is ten times too small, from reading the coefficient as 1.9×10^{-6} . The exponent on the coefficient is easy to misread, and the order-of-magnitude check in step 5 catches it: two metres of brass over a hundred degrees gives millimetres, not tenths of a millimetre.

The trap. Putting the final temperature into the formula. It is the number that appears in the sentence, so it is the number that gets used. Writing $\Delta T = 120 - 20$ as a separate line is the cheapest possible guard, and it costs a few seconds.

Relevant topics: linear thermal expansion; the coefficient of linear expansion; temperature change against temperature reading; why expansion joints exist; the order of magnitude of thermal expansion in metals.

The trap. Using the final temperature of 120 degrees instead of the 100 degree rise. The formula needs the CHANGE in temperature, and the starting temperature is given precisely so that it has to be subtracted.

Q23 · Mass-energy equivalence: the energy released by a mass defect — **Answer B**

What is being tested. A single relation, plus the discipline of handling powers of ten. There is no physics beyond $E = m c^2$, so the mark turns on the arithmetic.

Step 1 — why a mass defect gives energy. Mass and energy are two forms of the same thing, related by $E = m c^2$. When a reaction produces less mass than it started with, the missing mass has not vanished: it has been converted into energy and carried away by the products. So the mass to use is the *decrease*, not either total.

$$E = m c^2$$

Step 2 — square the speed of light. Do this as a separate line, because it is where the powers of ten get mangled:

$$\begin{aligned} c^2 &= (3.0 \times 10^8)^2 \\ &= 9.0 \times 10^{16} \text{ m}^2/\text{s}^2 \end{aligned}$$

Step 3 — multiply.

$$\begin{aligned} E &= 3.6 \times 10^{-30} \times 9.0 \times 10^{16} \\ &= 32.4 \times 10^{-14} \\ &= 3.2 \times 10^{-13} \text{ J} \end{aligned}$$

So **Answer: B**.

Step 4 — check the units. A kilogram times a square metre per square second is $\text{kg m}^2 \text{s}^{-2}$, which is a joule. The units work out, so the relation has been applied in the right form. A dimensional check on this relation is worth doing because the numbers are so far from everyday experience that intuition cannot help.

Step 5 — check the size. $3.2 \times 10^{-13} \text{ J}$ is about two million electronvolts, which is exactly the scale of energy released in a single nuclear transition. For one nucleus that is vanishingly small in everyday terms; for a mole of nuclei it is $3.2 \times 10^{-13} \times 6 \times 10^{23} = 2 \times 10^{11} \text{ J}$, which is the output of a large power station for a minute. The huge factor between one nucleus and a mole is why nuclear energy is worth harnessing at all, and it is a useful thing to carry in mind as a sanity check.

The distractors.

- **$1.1 \times 10^{-21} \text{ J}$** is $3.6 \times 10^{-30} \times 3.0 \times 10^8$: the speed of light used unsquared. The square is the whole content of the relation, and dropping it makes the answer smaller by a factor of 3×10^8 .
- **$4.0 \times 10^{-47} \text{ J}$** is $3.6 \times 10^{-30}/9.0 \times 10^{16}$: divided by c^2 instead of multiplied. It is the same confusion as the previous option but with the exponent's sign flipped, and it is a very small number — smaller than any single nuclear energy, which is a reason to distrust it.
- **$2.9 \times 10^4 \text{ J}$** is $3.6 \times 10^{-30} \times 8.1 \times 10^{33}$: c^4 rather than c^2 . It is enormous for one nucleus, and no single nuclear reaction releases tens of thousands of joules.
- **$3.2 \times 10^{-14} \text{ J}$** is exactly ten times too small, from reading the mass defect as 3.6×10^{-31} . It is the right answer to a slightly different question, which is the hardest kind of distractor to spot.

The trap. Losing track of the exponents. Multiplying 10^{-30} by 10^{16} is easy; the mistake comes earlier, in squaring 3.0×10^8 and getting 9.0×10^{16} rather than 9.0×10^8 or 6.0×10^{16} .

Squaring the coefficient and doubling the exponent as two separate operations is what keeps it straight.

Relevant topics: mass-energy equivalence; the mass defect and binding energy; powers of ten; the electronvolt as a nuclear energy unit; why the energy of one nucleus is tiny and of a mole is not.

The trap. Dropping the square on c , or squaring it twice. The relation is $E = mc^2$ with $c^2 = 9.0 \times 10^{16}$, and keeping the exponent on that one line is the whole of the arithmetic.

Q24 · The photoelectric effect: the stopping potential for a given wavelength — **Answer E**

What is being tested. The photoelectric equation applied to the energy of the electron rather than to the light, and the conversion from energy to potential that the electronvolt makes trivial.

Step 1 — the photon energy. Photon energy is hf , and since $f = c/\lambda$ it is also hc/λ . The combination hc is a constant, and in the units that make this question easy it is:

$$hc = 1240 \text{ eV nm}$$

which is worth carrying in mind, because it turns a wavelength in nanometres straight into an energy in electronvolts with one division:

$$hf = 1240/400 = 3.1 \text{ eV}$$

Step 2 — the photoelectric equation. An electron needs the work function just to escape, and whatever is left over becomes its kinetic energy:

$$\begin{aligned} hf &= \phi + E_{k,\max} \\ E_{k,\max} &= hf - \phi \\ &= 3.1 - 2.0 \\ &= 1.1 \text{ eV} \end{aligned}$$

The subtraction is not optional. A photon with 3.1 eV cannot give an electron 3.1 eV of motion, because 2.0 eV has already been spent on getting it out of the metal.

Step 3 — from energy to stopping potential. To stop an electron of kinetic energy E_k you must do work E_k on it. The work done in moving an electron through a potential difference V is eV , and the electronvolt is *defined* as the work done on one electron by one volt. So the numbers coincide:

$$\begin{aligned} eV_{\text{stop}} &= E_{k,\max} \\ V_{\text{stop}} &= E_{k,\max} \text{ in electronvolts} = 1.1 \text{ V} \end{aligned}$$

So **Answer: E**.

Step 4 — check the size. The photon energy is 3.1 eV and the work function 2.0 eV, so the electron can keep 1.1 eV — about a third of what arrived. The stopping potential of 1.1 V is a typical laboratory value for a photocell, and it is worth noting that it is less than the photon energy, which is always true and is a quick check on any answer of this kind.

Step 5 — check what the answer does not depend on. The stopping potential is a property of the *light* and the metal together, and not of how intense the light is. Brighter light of the same wavelength emits more electrons per second, but every one of them has the same maximum kinetic energy, so the

stopping potential is unchanged. That is one of the facts that the wave picture of light cannot explain and the photon picture can.

The distractors.

- **3.1 V** is the photon energy in electronvolts. It is the first number computed, and stopping there is the most natural error in the question: the work function has been ignored, so the electron is credited with energy it spent escaping.
- **5.1 V** adds the work function instead of subtracting it. It gives the electron more energy than the photon brought, which is impossible, and a moment's thought about conservation rules it out without any arithmetic.
- **2.0 V** is the work function. It is a real number in this problem and it is in electronvolts, so it looks like a legitimate answer — but it is a property of the metal, and the question is about the light.
- **0.55 V** is half the correct value. It would be right if only half the available energy reached the electron, and the photoelectric equation has no such halving in it.

The trap. Answering with the photon energy. The question gives a wavelength and asks for a voltage, and the photon energy sits in the middle of the working as an intermediate. Intermediates are what the wrong options are made of, and the discipline is to keep going until the quantity named in the question has been produced.

Relevant topics: the photoelectric effect; the work function; photon energy and wavelength; the stopping potential; why intensity changes the current but not the stopping potential.

The trap. Answering with the photon energy. The work function has to come off it, and the stopping potential is the leftover kinetic energy in electronvolts -- always less than the photon energy.

Q25 · Upthrust: what a spring balance reads with the load under water — **Answer D**

What is being tested. Whether you identify the forces on the stone and combine them with the right signs. The arithmetic is one subtraction; the whole question is the sign.

Step 1 — the two forces. With the stone hanging at rest under water, exactly two forces act on it: its weight downwards and the upthrust from the water upwards. Since it is not accelerating, the balance must supply whatever is needed to make up the difference. The balance therefore reads the *resultant*:

$$\text{reading} = \text{weight} - \text{upthrust}$$

Fixing the sign first is worth doing, because a sign error produces an answer that is physically impossible and easy to spot if you look for it.

Step 2 — the true weight.

$$W = m g = 0.50 \times 10 = 5.0 \text{ N}$$

Step 3 — the upthrust. Archimedes' principle: the upthrust equals the weight of the fluid displaced. The stone is completely submerged, so it displaces its own volume of water:

$$\begin{aligned}
 \text{upthrust} &= \rho_{\text{water}} \times V \times g \\
 &= 1000 \times 2.0 \times 10^{-4} \times 10 \\
 &= 2.0 \text{ N}
 \end{aligned}$$

Notice that the stone's own mass does not appear in this line at all. The upthrust is fixed by how much water is pushed out of the way, and a stone of the same volume but twice the mass would feel exactly the same 2.0 N.

Step 4 — combine.

$$\text{reading} = 5.0 - 2.0 = 3.0 \text{ N}$$

So **Answer: D.**

Step 5 — check the direction and the size. The reading must be less than the true weight, because the water pushes up and helps. It must be greater than zero, because the stone sinks. **3.0 N** satisfies both: it is less than 5.0 N and more than nothing. Any answer above 5.0 N, or negative, or exactly zero, fails one of these bounds without any calculation — and two of the options do exactly that.

Step 6 — confirm that the stone really does sink. Its density is $0.50 / (2.0 \times 10^{-4}) = 2500 \text{ kg/m}^3$, which is two and a half times the density of water. So the weight exceeds the upthrust by a factor of two and a half, and the stone sinks. That is consistent with the reading of 3.0 N being positive. A stone of the same volume with a mass below 0.20 kg would float, and the balance would read zero.

The distractors.

- **5.0 N** is the stone's true weight. It is what the balance reads in air, and it is the answer if the upthrust is forgotten. This is the commonest error, and the whole point of the question is that weighing under water gives a smaller number.
- **2.0 N** is the upthrust by itself. It is a force in the problem and it is in newtons, so it looks like a candidate — but the balance does not read the upthrust, it reads the resultant of the upthrust and the weight.
- **7.0 N** adds the two forces. That would mean the stone is *heavier* under water than in air, which contradicts the everyday experience of lifting something in a bath. The sign check in step 5 rules it out immediately.
- **0 N** is the reading for a floating object, where the upthrust exactly balances the weight. This stone has a density of 2500 kg/m^3 against water's 1000, so it cannot float; the balance would read zero only if the stone's mass were 0.20 kg or less.

The trap. Using the stone's weight to compute the upthrust. The upthrust depends on the volume of water displaced, and the volume is given separately for exactly this reason. A stone twice as dense but the same size would feel the same upthrust, and a stone twice the volume would feel twice as much.

Relevant topics: Archimedes' principle; upthrust and displaced fluid; density and floating; combining two forces with opposite signs; the apparent weight of a submerged body.

The trap. Forgetting the upthrust, or adding it instead of subtracting it. The balance reads the resultant of the weight and the upthrust, so the reading must be less than the true weight and greater than zero.